

Reconstruction Metric - GPU Optimized

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0.1 1. Setup e Imports

```
import numpy as np
import pandas as pd
import sys
import os
import subprocess
import time
import torch
import torch.nn as nn
import torch.nn.functional as F
import matplotlib.pyplot as plt

from pathlib import Path
from tqdm import tqdm
from sae.models.sae import SparseAutoencoder
from sae.tools.experiment_utils import get_experiment_dir, get_metrics_dir, get_report_name
from sae.tools.naming_utils import get_checkpoint_dir, get_model_name, get_extras_id

project_root = Path('...').resolve()
sys.path.insert(0, str(project_root))

device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
print(f"Device: {device}")
if torch.cuda.is_available():
    print(f"GPU: {torch.cuda.get_device_name(0)}")
    print(f"Memoria: {torch.cuda.get_device_properties(0).total_memory / 1e9:.2f} GB")
```

```
Device: cuda
GPU: NVIDIA GeForce RTX 5060
Memoria: 8.55 GB
```

```
# Configuración para visualización en PDF
pd.set_option('display.max_colwidth', None)
pd.set_option('display.max_columns', None)
pd.set_option('display.width', 100)
```

```

np.set_printoptions(linewidth=100, edgeitems=3)

os.environ['COLUMNS'] = '100'

print(" Configuración para PDF lista")

```

Configuración para PDF lista

```

NAMING_META= {
'variante': 'standard', #ArquitecturadelSAE
'tecnica': 'p-annealing', #P-annealing(Lp~pconpdecreciente)
'capa':6, #LayerdelmodeloOthelloGPT
'juegos': 5000, #Número departidas
'base_path': 'C:\\Users\\Esposa\\Documents\\Repos\\sae-othello-gpt', #Pathbaseparacheckpoints
'experiment_base_path':os.path.abspath('../../experiments'), #sae/experiments/
'extras':{
'expansion':16,
'sparsity_coef': 5e-1,
'epochs':200,
'early_stopping': False,
'learning_rate': 1e-3,
'p_start': 1.0,
'p_end': 0.2,
'n_sparsity_updates': 10,
'anneal_start':2000,
'batch_size': 2048,
'warmup_steps':400,
'sparsity_warmup_steps': 800,
'estimated_epochs': 200
}
}
print('\n Metadata de naming:')
for key, value in NAMING_META.items():
    print(f' {key}: {value}')

```

Metadata de naming:

```

variante: standard
tecnica: p-annealing
capa: 6
juegos: 5000
base_path: C:\Users\Esposa\Documents\Repos\sae-othello-gpt
experiment_base_path: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments
extras: {'expansion': 16, 'sparsity_coef': 0.5, 'epochs': 200, 'early_stopping': False, 'learning_rate': 0.001, 'p_start': 1.0, 'p_end': 0.2, 'n_sparsity_updates': 10, 'anneal_start': 2000, 'batch_size': 2048, 'warmup_steps': 400, 'sparsity_warmup_steps': 800, 'estimated_epochs': 200}

```

0.2 2. Cargar Datos

```

# Cargar activaciones
activations_path = project_root / "activations" / "data" / "layer6_2000games.npy"
activations = np.load(activations_path)

print(f"Activaciones del modelo:")
print(f" Shape: {activations.shape}")
print(f" Memoria: {activations.nbytes / (1024**2):.2f} MB")

```

Activaciones del modelo:

```

Shape: (118000, 512)
Memoria: 230.47 MB

```

```
# Cargar ground truth de BSPs
bsp_gt_path = project_root / "metrics" / "02_data" / "bsp_ground_truth_2000games.npy"
bsp_names_path = project_root / "metrics" / "02_data" / "bsp_ground_truth_2000games.names.npy"

bsp_ground_truth = np.load(bsp_gt_path)
bsp_names = np.load(bsp_names_path, allow_pickle=True)

print(f"\nGround truth BSPs:")
print(f"  Shape: {bsp_ground_truth.shape}")
print(f"  Total BSPs: {len(bsp_names)}")
```

Ground truth BSPs:
 Shape: (118000, 326)
 Total BSPs: 326

0.3 3. Cargar SAE y Extraer Features

```
# Configuración del SAE
input_dim = 512
hidden_dim = 8192

# Construir ruta del modelo usando naming_utils
checkpoint_dir = get_checkpoint_dir(
    NAMING_META['base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos']
)
extras_id = get_extras_id(checkpoint_dir, NAMING_META['extras']) if NAMING_META.get('extras') else None

model_filename = get_model_name(
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    'final',
    extras_id=extras_id
)
model_path = checkpoint_dir / model_filename

# Cargar modelo SAE estándar
sae = SparseAutoencoder(input_dim, hidden_dim).to(device)

checkpoint = torch.load(model_path, map_location=device, weights_only=False)
sae.load_state_dict(checkpoint['model_state_dict'])
sae.eval()

print(f"SAE cargado:")
print(f"  Path: {model_path}")
print(f"  Input: {input_dim}, Hidden: {hidden_dim}")
print(f"  Expansion: {hidden_dim/input_dim}x")
```

SAE cargado:
 Path: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\sae_p-annealing_standard\5000games_annealing_16_5000g_ex2_final.pt
 Input: 512, Hidden: 8192
 Expansion: 16.0x

```
# Extraer features del SAE en GPU
print("Extrayendo features del SAE...")
activations_tensor = torch.from_numpy(activations).float().to(device)

with torch.no_grad():
    sae_features = sae.encode(activations_tensor)

print(f"\nFeatures SAE:")
print(f"  Shape: {sae_features.shape}")
print(f"  Device: {sae_features.device}")
sparsity = (sae_features == 0).float().mean().item()
print(f"  Sparsity: {sparsity:.2%}")
print(f"  Activaciones promedio: {sae_features.shape[1] * (1 - sparsity):.1f}")
```

Extrayendo features del SAE...

```
Features SAE:
  Shape: torch.Size([118000, 8192])
  Device: cuda:0
  Sparsity: 98.27%
  Activaciones promedio: 141.4
```

```
del activations_tensor
torch.cuda.empty_cache()
```

0.4 4. Split Train/Test y Filtrar BSPs de Piezas

```
# Split 50/50:
n_moves = 59
n_games = 2000 # número de partidas para prueba de métricas
train_games = n_games // 2
split_idx = train_games * n_moves

sae_features_train = sae_features[:split_idx]
sae_features_test = sae_features[split_idx:]

print(f"Total partidas: {n_games}")
print(f"Train: {train_games} partidas ({split_idx} activaciones)")
print(f"Test: {n_games - train_games} partidas ({len(activations) - split_idx} activaciones)")
print(f"Train set: {sae_features_train.shape}")
print(f"Test set: {sae_features_test.shape}")
```

```
Total partidas: 2000
Train: 1000 partidas (59000 activaciones)
Test: 1000 partidas (59000 activaciones)
Train set: torch.Size([59000, 8192])
Test set: torch.Size([59000, 8192])
```

```
# =====
# CONFIGURACIÓN DE FILTRADO DE BSPs
# =====
USE_PLAYER = False # BSPs perspectiva del jugador (1 y 2)
USE_ABSOLUTE = True # BSPs absolutas negro/blanco (B y W)
USE_STRATEGIC = False # BSPs especiales (BSP_)

bsp_piezas_indices = []
bsp_piezas_names = []

for i, name in enumerate(bsp_names):
```

```

include = False
if USE_PLAYER and len(name) == 6 and name.startswith('BSP') and (name.endswith('1') or name.endswith('2')):
    include = True
if USE_ABSOLUTE and (name.endswith('B') or name.endswith('W')):
    include = True
if USE_STRATEGIC and name.startswith('BSP_'):
    include = True
if include:
    bsp_pieces_indices.append(i)
    bsp_pieces_names.append(name)

bsp_pieces_indices = np.array(bsp_pieces_indices)

# Para reconstruction necesitas split train/test
bsp_gt_train_pieces = torch.from_numpy(bsp_ground_truth[:split_idx, bsp_pieces_indices]).bool().to(device)
bsp_gt_test_pieces = torch.from_numpy(bsp_ground_truth[split_idx:, bsp_pieces_indices]).bool().to(device)

# Convertir a tensor en GPU
# Identificador del filtrado (para naming de archivos de salida)
parts = []
if USE_PLAYER: parts.append("player")
if USE_ABSOLUTE: parts.append("absolute")
if USE_STRATEGIC: parts.append("strategic")
filter_suffix = "_".join(parts) if parts else "none"

print(f"BSPs seleccionadas para Reconstruction:")
print(f"  Player: {USE_PLAYER} | Absolute: {USE_ABSOLUTE} | Strategic: {USE_STRATEGIC}")
print(f"  Total: {len(bsp_pieces_indices)}")
print(f"  Train shape: {bsp_gt_train_pieces.shape}")
print(f"  Test shape: {bsp_gt_test_pieces.shape}")
print(f"  Device: {bsp_gt_train_pieces.device}")
print(f"  Primeras 5: {'', '.join(bsp_pieces_names[:5])}")
print(f"  Últimas 5: {'', '.join(bsp_pieces_names[-5:])}")
print(f"  Filter suffix: {filter_suffix}")

```

```

BSPs seleccionadas para Reconstruction:
Player: False | Absolute: True | Strategic: False
Total: 128
Train shape: torch.Size([59000, 128])
Test shape: torch.Size([59000, 128])
Device: cuda:0
Primeras 5: BSPA1B, BSPA1W, BSPA2B, BSPA2W, BSPA3B
Últimas 5: BSPH6W, BSPH7B, BSPH7W, BSPH8B, BSPH8W
Filter suffix: absolute

```

0.5 5. Funciones GPU-Optimizadas

```

def fast_precision_score_gpu(y_true, y_pred, eps=1e-8):
    """
    Precisión vectorizada en GPU.

    Args:
        y_true: Tensor (n_positions,) booleano
        y_pred: Tensor (n_positions, n_candidates) booleano

    Returns:
        precision_scores: Tensor (n_candidates,)
    """
    y_true_expanded = y_true.unsqueeze(1)

```

```

tp = (y_true_expanded & y_pred).sum(dim=0).float()
fp = (~y_true_expanded & y_pred).sum(dim=0).float()

precision = tp / (tp + fp + eps)

return precision

def fast_f1_score_gpu(y_true, y_pred, eps=1e-8):
    """
    F1 score vectorizado en GPU.

    Args:
        y_true: Tensor (n_positions,) booleano
        y_pred: Tensor (n_positions,) booleano

    Returns:
        f1: Float
    """
    tp = (y_true & y_pred).sum().float()
    fp = (~y_true & y_pred).sum().float()
    fn = (y_true & ~y_pred).sum().float()

    precision = tp / (tp + fp + eps)
    recall = tp / (tp + fn + eps)

    f1 = 2 * (precision * recall) / (precision + recall + eps)

    return f1.item()

print(" Funciones GPU definidas")

```

Funciones GPU definidas

0.6 6. Fase 1: Identificar Features de Alta Precisión (GPU)

Para cada BSP, encontrar features con precisión 0.95 en el train set.

```

def identify_high_precision_features_gpu(
    sae_features_train,
    bsp_gt_train,
    f_max,
    precision_threshold=0.95,
    significance_threshold=10,
    thresholds=[0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9],
    feature_batch_size=512,
    verbose=True
):
    """
    Para cada (threshold t, feature f, BSP b): determina si f es clasificador
    de alta precisión para b al threshold t.

    Criterios (igual que implementación oficial):
        - precision(f, b, t) >= precision_threshold (default 0.95)
        - on_count(f, t) >= significance_threshold (default 10)

    Args:
        sae_features_train: Tensor (P, n_features) GPU
        bsp_gt_train: Tensor (P, n_bsps) GPU, bool
        f_max: Tensor (n_features,) GPU - calculado FUERA para
        garantizar identidad exacta con Fase 2
    """

```

```

precision_threshold: float, default 0.95
significance_threshold: int, mín. activaciones requeridas (default 10)
thresholds: lista de fracciones t [0, 1)
feature_batch_size: int, features por mini-batch interno

Returns:
    hp_mask_TFB: BoolTensor (T, n_active, n_bsps)
    active_feature_indices: LongTensor (n_active,)
"""
n_positions, n_features = sae_features_train.shape
n_bsps = bsp_gt_train.shape[1]
device = sae_features_train.device

thresholds_T = torch.tensor(thresholds, device=device)
n_thresholds = len(thresholds_T)

# Features vivas (f_max > 0)
active_feature_indices = (f_max > 0).nonzero(as_tuple=True)[0]
n_active = len(active_feature_indices)

if verbose:
    print("=" * 60)
    print("FASE 1: IDENTIFICAR FEATURES DE ALTA PRECISIÓN")
    print("=" * 60)
    print(f"Precision threshold: {precision_threshold}")
    print(f"Significance threshold: >= {significance_threshold} activaciones")
    print(f"Features activas: {n_active}/{n_features} ({n_active/n_features*100:.1f}%)")
    print(f"BSPs: {n_bsps}")
    print(f"Thresholds: {thresholds}")
    print("=" * 60)

# Thresholds absolutos: thresh_TF[t, f] = t * f_max[f]
active_f_max = f_max[active_feature_indices] # (n_active,)
thresholds_TF = thresholds_T[:, None] * active_f_max # (T, n_active)

# Acumuladores sobre todo el train set
on_count_TF = torch.zeros(n_thresholds, n_active, device=device)
tp_count_TFB = torch.zeros(n_thresholds, n_active, n_bsps, device=device)

bsp_float_PB = bsp_gt_train.float() # (P, B)

n_iters = (n_active + feature_batch_size - 1) // feature_batch_size

for fi in tqdm(range(n_iters), desc="Fase 1"):
    f_start = fi * feature_batch_size
    f_end = min(f_start + feature_batch_size, n_active)

    global_idx = active_feature_indices[f_start:f_end]
    acts_PF = sae_features_train[:, global_idx] # (P, bs)
    thresh_TF = thresholds_T[:, f_start:f_end] # (T, bs)

    # active_TFP[t, f, p] = True si acts[p,f] > thresh[t,f]
    active_TFP = acts_PF.T.unsqueeze(0) > thresh_TF.unsqueeze(2) # (T, bs, P)

    # Conteo de activaciones por (threshold, feature)
    on_count_TF[:, f_start:f_end] += active_TFP.sum(dim=2).float()

    # True positives: f activa Y BSP verdadera
    # tp[t, f, b] = Σ_p active[t,f,p] * bsp[p,b]
    tp_count_TFB[:, f_start:f_end, :] += torch.einsum(
        'tfp,pb->tfb', active_TFP.float(), bsp_float_PB
    )

```

```

# Precisión: tp / on_count
eps = 1e-8
precision_TFB = tp_count_TFB / (on_count_TF.unsqueeze(2) + eps) # (T, F, B)

# Máscara: alta precisión AND significancia suficiente
sig_mask_TF1 = (on_count_TF >= significance_threshold).unsqueeze(2) # (T, F, 1)
hp_mask_TFB = (precision_TFB >= precision_threshold) & sig_mask_TF1 # (T, F, B)

if verbose:
    hp_per_t = hp_mask_TFB.sum(dim=(1, 2)) # (T,)
    hp_per_b = hp_mask_TFB.sum(dim=1) # (T, B)
    best_t = hp_per_t.argmax().item()
    print(f"\nPares (feature, BSP) de alta precisión por threshold:")
    for i, t in enumerate(thresholds):
        marker = " ← más pares" if i == best_t else ""
        print(f"  t={t:.1f}: {hp_per_t[i].item():5d} pares{marker}")
    print(f"\nFeatures HP por BSP en t={thresholds[best_t]:.1f}:")
    print(f"  Media: {hp_per_b[best_t].float().mean():.1f}")
    print(f"  Máx: {hp_per_b[best_t].max().item()}")
    print(f"  BSPs sin features: {(hp_per_b[best_t] == 0).sum().item()}")

    return hp_mask_TFB, active_feature_indices

print(" Fase 1 rediseñada: umbral de significancia (10) + estructura (T, F, B)")

```

Fase 1 rediseñada: umbral de significancia (10) + estructura (T, F, B)

```

THRESHOLDS = torch.arange(0.0, 1.05, 0.05).tolist()

# f_max calculado UNA SOLA VEZ aquí - se pasa a Fase 1 y Fase 2
f_max = sae_features_train.max(dim=0)[0] # (n_features,)
print(f"f_max calculado: {(f_max > 0).sum().item()} features activas de {f_max.shape[0]}")

start_time = time.time()

hp_mask_TFB, active_feature_indices = identify_high_precision_features_gpu(
    sae_features_train,
    bsp_gt_train_pieces,
    f_max,
    precision_threshold=0.95,
    significance_threshold=10,
    thresholds=THRESHOLDS,
    feature_batch_size=512,
    verbose=True
)

phase1_time = time.time() - start_time

print(f"\nTiempo Fase 1: {phase1_time:.2f} seg")
print(f"hp_mask_TFB shape: {hp_mask_TFB.shape} (T, n_active, n_bsps)")

```

f_max calculado: 8192 features activas de 8192

FASE 1: IDENTIFICAR FEATURES DE ALTA PRECISIÓN

Precision threshold: 0.95

Significance threshold: >= 10 activaciones

Features activas: 8192/8192 (100.0%)

BSPs: 128

Thresholds: [0.0, 0.05000000074505806, 0.10000000149011612, 0.15000000596046448, 0.20000000298023224, 0.25000000596046448, 0.30000000796046448, 0.35000001196046448, 0.40000001596046448, 0.45000001996046448, 0.50000002396046448, 0.55000002796046448, 0.60000003196046448, 0.65000003596046448, 0.70000003996046448, 0.75000004396046448, 0.80000004796046448, 0.85000005196046448, 0.90000005596046448, 0.95000005996046448, 1.00000006396046448]

Fase 1: 100% | 16/16 [01:05<00:00, 4.10s/it]

Pares (feature, BSP) de alta precisión por threshold:

```
t=0.0: 0 pares
t=0.1: 19 pares
t=0.1: 69 pares
t=0.2: 175 pares
t=0.2: 347 pares
t=0.2: 515 pares
t=0.3: 771 pares
t=0.3: 1010 pares
t=0.4: 1274 pares
t=0.5: 1454 pares
t=0.5: 1547 pares
t=0.6: 1581 pares
t=0.6: 1589 pares ← más pares
t=0.6: 1544 pares
t=0.7: 1421 pares
t=0.8: 1365 pares
t=0.8: 1228 pares
t=0.9: 1105 pares
t=0.9: 991 pares
t=0.9: 938 pares
t=1.0: 0 pares
```

Features HP por BSP en t=0.6:

```
Media: 12.4
Máx: 177
BSPs sin features: 56
```

Tiempo Fase 1: 80.31 seg

hp_mask_TFB shape: torch.Size([21, 8192, 128]) (T, n_active, n_bsps)

0.7 7. Fase 2: Reconstruir Tableros en Test Set (GPU)

```
def reconstruct_boards_gpu(
    sae_features_test,
    bsp_gt_test,
    hp_mask_TFB,
    active_feature_indices,
    f_max,
    thresholds=[0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9],
    verbose=True
):
    """
    Fase 2: barrido simultáneo de thresholds T → toma el mejor F1.

    Para cada threshold t:
    1. Binarizar features de test: active[p, f] = acts[p, f] > t * f_max[f]
    2. Predicción por (posición, BSP):
        predict[p, b] = OR_f ( active[p,f] AND hp_mask[t, f, b] )
        implementado como: (active_PF @ hp_FB) > 0
    3. F1 global micro-averaged (acumula TP/FP/FN sobre todas las posiciones y BSPs)

    Retorna el max F1 sobre todos los thresholds (Eq. 5 del paper: max_t).

    Args:
        sae_features_test: Tensor (P, n_features) GPU
        bsp_gt_test: Tensor (P, n_bsps) GPU, bool
```

```

hp_mask_TFB:          BoolTensor (T, n_active, n_bsps) de Fase 1
active_feature_indices: LongTensor (n_active,) de Fase 1
f_max:                Tensor (n_features,) GPU - MISMO objeto que Fase 1
thresholds:           lista de fracciones (debe coincidir con Fase 1)

Returns:
    best_f1:           float, max F1 sobre thresholds
    best_threshold:    float, threshold óptimo
    f1_per_threshold:  np.array (T,), F1 por threshold
"""
device = sae_features_test.device
n_thresholds = len(thresholds)
thresholds_T = torch.tensor(thresholds, device=device)

# Activaciones del test solo para features activas
test_active_PF = sae_features_test[:, active_feature_indices] # (P, n_active)
active_f_max = f_max[active_feature_indices] # (n_active,)

gt_PB = bsp_gt_test.bool() # (P, B)
eps = 1e-8

f1_scores = torch.zeros(n_thresholds, device=device)
precision_T = torch.zeros(n_thresholds, device=device)
recall_T = torch.zeros(n_thresholds, device=device)

if verbose:
    print("=" * 60)
    print("FASE 2: RECONSTRUCCIÓN - BARRIDO DE THRESHOLDS")
    print("=" * 60)
    print(f"{'t':>5} {'F1':>8} {'P':>8} {'R':>8} {'TP':>8} {'FP':>8} {'FN':>8}")
    print("-" * 60)

for t_idx in range(n_thresholds):
    t = thresholds[t_idx]

    # Threshold absoluto por feature: t * f_max[f]
    abs_thresh_F = thresholds_T[t_idx] * active_f_max # (n_active,)

    # Binarizar features en test
    active_PF = test_active_PF > abs_thresh_F.unsqueeze(0) # (P, n_active)

    # predict[p, b] = any( active[p,f] AND hp[t,f,b] )
    # = (active_PF.float() @ hp_mask[t].float()) > 0
    hp_FB = hp_mask_TFB[t_idx].float() # (n_active, B)
    predictions_PB = (active_PF.float() @ hp_FB) > 0 # (P, B)

    # Métricas globales micro-averaged
    tp = (predictions_PB & gt_PB).sum().float()
    fp = (predictions_PB & ~gt_PB).sum().float()
    fn = (~predictions_PB & gt_PB).sum().float()

    p = tp / (tp + fp + eps)
    r = tp / (tp + fn + eps)
    f1 = 2 * p * r / (p + r + eps)

    f1_scores[t_idx] = f1
    precision_T[t_idx] = p
    recall_T[t_idx] = r

    if verbose:
        print(f"{'t':>5.1f} {'f1.item():>8.4f} {'p.item():>8.4f} {'r.item():>8.4f}"
              f" {'tp.item():>8.0f} {'fp.item():>8.0f} {'fn.item():>8.0f}")

```

```

best_idx      = f1_scores.argmax().item()
best_f1       = f1_scores[best_idx].item()
best_threshold = thresholds[best_idx]

if verbose:
    print("=" * 60)
    print(f"Mejor threshold: t={best_threshold:.1f}")
    print(f"Reconstruction Score (max_t F1): {best_f1:.4f}")
    print("=" * 60)

return best_f1, best_threshold, f1_scores.cpu().numpy()

print(" Fase 2 rediseñada: barrido T simultáneo + F1 global micro-averaged + max_t")

```

Fase 2 rediseñada: barrido T simultáneo + F1 global micro-averaged + max_t

```

start_time = time.time()

reconstruction_score, best_threshold, f1_per_threshold = reconstruct_boards_gpu(
    sae_features_test,
    bsp_gt_test_pieces,
    hp_mask_TFB,
    active_feature_indices,
    f_max,          # mismo tensor calculado antes de Fase 1
    thresholds=THRESHOLDS,
    verbose=True
)

phase2_time = time.time() - start_time
total_time  = phase1_time + phase2_time

print(f"\nTiempo Fase 2: {phase2_time:.2f} seg")
print(f"Tiempo total:  {total_time:.2f} seg ({total_time/60:.2f} min)")

```

```

=====
FASE 2: RECONSTRUCCIÓN - BARRIDO DE THRESHOLDS
=====

```

t	F1	P	R	TP	FP	FN
0.0	0.0000	0.0000	0.0000	0	0	2006000
0.1	0.0045	0.9705	0.0023	4567	139	2001433
0.1	0.0105	0.9784	0.0053	10544	233	1995456
0.2	0.0164	0.9706	0.0082	16544	502	1989456
0.2	0.0219	0.9694	0.0111	22216	702	1983784
0.2	0.0247	0.9700	0.0125	25097	777	1980903
0.3	0.0282	0.9648	0.0143	28657	1044	1977343
0.3	0.0303	0.9657	0.0154	30843	1096	1975157
0.4	0.0318	0.9688	0.0162	32422	1043	1973578
0.5	0.0316	0.9727	0.0161	32204	905	1973796
0.5	0.0310	0.9738	0.0158	31633	852	1974367
0.6	0.0302	0.9726	0.0153	30744	867	1975256
0.6	0.0287	0.9737	0.0146	29196	788	1976804
0.6	0.0269	0.9789	0.0137	27388	590	1978612
0.7	0.0248	0.9825	0.0126	25191	449	1980809
0.8	0.0235	0.9840	0.0119	23886	388	1982114
0.8	0.0218	0.9886	0.0110	22077	254	1983923
0.9	0.0205	0.9926	0.0104	20815	156	1985185
0.9	0.0194	0.9945	0.0098	19636	109	1986364
0.9	0.0187	0.9974	0.0094	18947	50	1987053
1.0	0.0000	0.0000	0.0000	0	0	2006000

```

=====

```

```
Mejor threshold: t=0.4
Reconstruction Score (max_t F1): 0.0318
=====
```

```
Tiempo Fase 2: 17.58 seg
Tiempo total: 97.89 seg (1.63 min)
```

0.8 8. Análisis de Resultados

```
print("=" * 60)
print("BARRIDO DE THRESHOLDS - F1 POR THRESHOLD")
print("=" * 60)
for i, (t, f1) in enumerate(zip(THRESHOLDS, f1_per_threshold)):
    marker = " ← MEJOR" if i == f1_per_threshold.argmax() else ""
    print(f"t={t:.1f}: F1={f1:.4f}{marker}")
print("=" * 60)

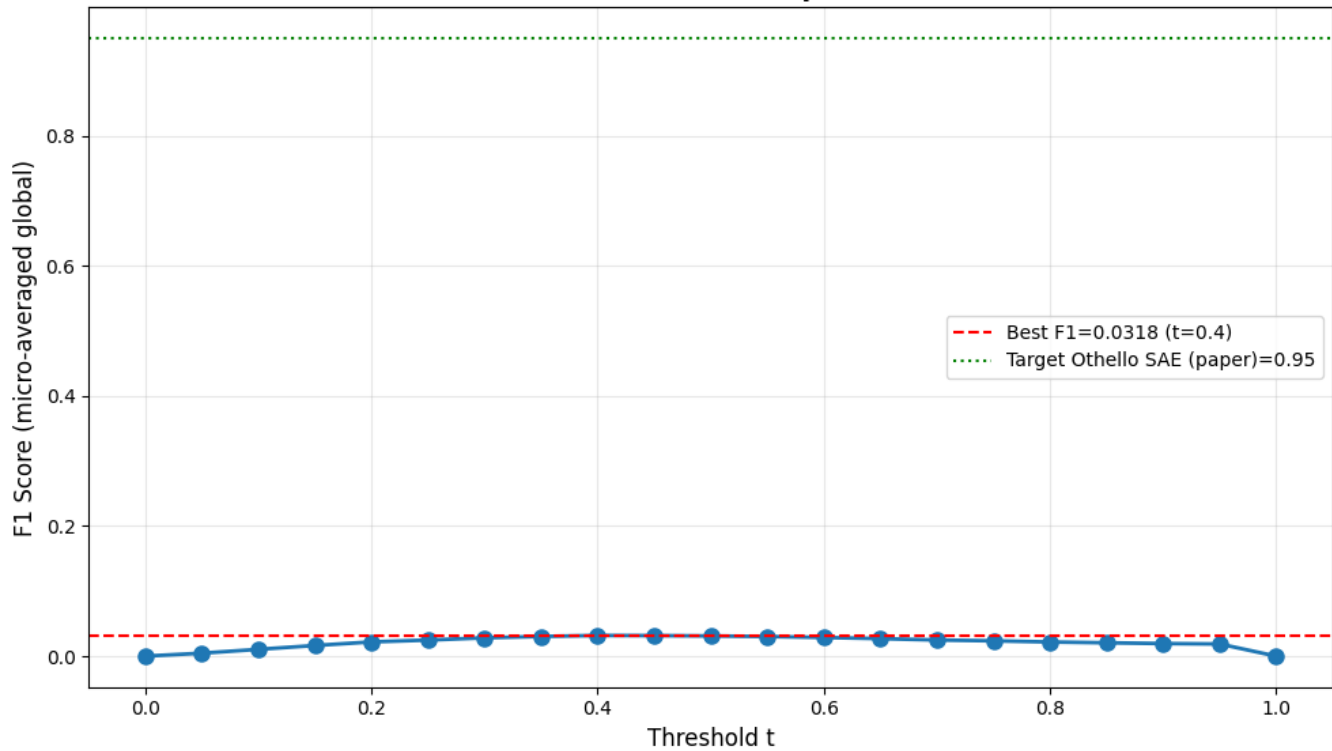
plt.figure(figsize=(10, 6))
plt.plot(THRESHOLDS, f1_per_threshold, marker='o', linewidth=2, markersize=8)
plt.axhline(y=reconstruction_score, color='r', linestyle='--',
            label=f'Best F1={reconstruction_score:.4f} (t={best_threshold:.1f})')
plt.axhline(y=0.95, color='g', linestyle=':', label='Target Othello SAE (paper)=0.95')
plt.xlabel('Threshold t', fontsize=12)
plt.ylabel('F1 Score (micro-averaged global)', fontsize=12)
plt.title('Board Reconstruction: F1 por Threshold', fontsize=14, fontweight='bold')
plt.grid(True, alpha=0.3)
plt.legend()
plt.tight_layout()
plt.show()

print(f"\n Threshold óptimo: t={best_threshold:.1f}")
print(f" Reconstruction Score: {reconstruction_score:.4f}")
```

```
=====
BARRIDO DE THRESHOLDS - F1 POR THRESHOLD
=====
```

```
t=0.0: F1=0.0000
t=0.1: F1=0.0045
t=0.1: F1=0.0105
t=0.2: F1=0.0164
t=0.2: F1=0.0219
t=0.2: F1=0.0247
t=0.3: F1=0.0282
t=0.3: F1=0.0303
t=0.4: F1=0.0318 ← MEJOR
t=0.5: F1=0.0316
t=0.5: F1=0.0310
t=0.6: F1=0.0302
t=0.6: F1=0.0287
t=0.6: F1=0.0269
t=0.7: F1=0.0248
t=0.8: F1=0.0235
t=0.8: F1=0.0218
t=0.9: F1=0.0205
t=0.9: F1=0.0194
t=0.9: F1=0.0187
t=1.0: F1=0.0000
=====
```

Board Reconstruction: F1 por Threshold



Threshold óptimo: $t=0.4$

Reconstruction Score: 0.0318

```
# Estadísticas sobre features de alta precisión por BSP
hp_per_bsp = hp_mask_TFB.sum(dim=1) # (T, n_bsps)
best_t_idx = torch.tensor(f1_per_threshold).argmax().item()
n_features_per_bsp = hp_per_bsp[best_t_idx].cpu().numpy()

print("="*60)
print("ESTADÍSTICAS - FEATURES POR BSP")
print("="*60)
print(f"Total BSPs: {len(n_features_per_bsp)}")
print(f"Features promedio por BSP: {np.mean(n_features_per_bsp):.1f}")
print(f"Features mediana por BSP: {np.median(n_features_per_bsp):.1f}")
print(f"BSPs con 0 features: {np.sum(n_features_per_bsp == 0)}")
print(f"BSPs con 1+ features: {np.sum(n_features_per_bsp > 0)}")
print(f"BSPs con 5+ features: {np.sum(n_features_per_bsp >= 5)}")
print(f"BSPs con 10+ features: {np.sum(n_features_per_bsp >= 10)}")
print(f"Max features en una BSP: {np.max(n_features_per_bsp)}")
print("="*60)
```

```
=====
ESTADÍSTICAS - FEATURES POR BSP
=====
Total BSPs: 128
Features promedio por BSP: 10.0
Features mediana por BSP: 0.0
BSPs con 0 features: 65
BSPs con 1+ features: 63
BSPs con 5+ features: 40
BSPs con 10+ features: 25
Max features en una BSP: 155
=====
```

```
# Comparación con resultados del paper
print("="*60)
print("COMPARACIÓN CON PAPER (Karvonen et al., NeurIPS 2024)")
print("="*60)
print(f"{'Modelo':<20} {'Reconstruction':<15}")
print("-"*60)
print(f"{'SAE random':<20} {'0.08':<15}")
print(f"{'SAE trained (paper)':<20} {'0.95':<15} ← Objetivo")
print(f"{'Linear probe':<20} {'0.99':<15}")
print(f"{'Nuestro SAE':<20} {'reconstruction_score:<15.4f'} ← Resultado")
print("="*60)

if reconstruction_score >= 0.90:
    print("\n Excelente: Muy cercano al objetivo del paper")
elif reconstruction_score >= 0.80:
    print("\n Bueno: Por encima de random pero debajo del objetivo")
else:
    print("\n Bajo: Revisar implementación o entrenamiento del SAE")
```

```
=====
COMPARACIÓN CON PAPER (Karvonen et al., NeurIPS 2024)
=====
Modelo                Reconstruction
-----
SAE random            0.08
SAE trained (paper)   0.95                ← Objetivo
Linear probe          0.99
Nuestro SAE           0.0318            ← Resultado
=====
```

Bajo: Revisar implementación o entrenamiento del SAE

0.9 9. Guardar Resultados

```
results = {
    'reconstruction_score': reconstruction_score,
    'best_threshold': best_threshold,
    'f1_per_threshold': f1_per_threshold,
    'thresholds': np.array(THRESHOLDS),
    'phase1_time_seconds': phase1_time,
    'phase2_time_seconds': phase2_time,
    'total_time_seconds': total_time,
    'bsp_names': bsp_pieces_names,
}

metrics_dir = get_metrics_dir(
    NAMING_META['experiment_base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    extras_id=extras_id
)
output_path = metrics_dir / f"reconstruction_{filter_suffix}.npz"
np.savez(output_path, **results)

print(f" Resultados guardados en: {output_path}")
print(f"\nContenido:")
print(f" reconstruction_score: {reconstruction_score:.4f}")
print(f" best_threshold: {best_threshold:.1f}")
```

```

print(f"  fl_per_threshold:      {fl_per_threshold}")

# Comparación con el paper
print()
print("=" * 60)
print("COMPARACIÓN CON PAPER (Karvonen et al., NeurIPS 2024)")
print("=" * 60)
print(f"{'Modelo':<25} {'Reconstruction':>14}")
print("-" * 40)
print(f"{'SAE random':<25} {'0.08':>14}")
print(f"{'SAE trained (paper)':<25} {'0.95':>14} ← Objetivo Othello")
print(f"{'Linear probe':<25} {'0.99':>14}")
print(f"{'Nuestro SAE':<25} {reconstruction_score:>14.4f} ← Resultado")
print("=" * 60)

if reconstruction_score >= 0.90:
    print("\n Excelente: Muy cercano al objetivo del paper")
elif reconstruction_score >= 0.50:
    print("\n~ Moderado: Por encima de random, revisar SAE")
else:
    print("\n Bajo: Revisar entrenamiento del SAE o calidad de las activaciones")

```

Resultados guardados en: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standard\sae_p-annealing_standard\5000games\ex2\metrics\reconstruction_absolute.npz

Contenido:

```

reconstruction_score: 0.0318
best_threshold:      0.4
fl_per_threshold:    [0.          0.00454268 0.01045629 0.01635554 0.02189936 0.02470331 0.02815443 0.03026882 0.0317
0.03158635 0.03103579 0.03017652 0.02867999 0.02693048 0.02479869 0.02352983 0.02176864 0.02053803
0.01938645 0.01871311 0.          ]

```

```

=====
COMPARACIÓN CON PAPER (Karvonen et al., NeurIPS 2024)
=====
Modelo                Reconstruction
-----
SAE random            0.08
SAE trained (paper)   0.95 ← Objetivo Othello
Linear probe          0.99
Nuestro SAE           0.0318 ← Resultado
=====

```

Bajo: Revisar entrenamiento del SAE o calidad de las activaciones

```

import subprocess
from pathlib import Path
from sae.tools.experiment_utils import get_report_name

output_dir = get_metrics_dir(
    NAMING_META['experiment_base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    extras_id=extras_id
)

pdf_name = get_report_name(
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],

```

```

NAMING_META['juegos'],
f'reconstruction_{filter_suffix}',
extras_id=extras_id
)

notebook_name = 'reconstruction_sae_standar2.ipynb'
notebook_abs = str(Path('.').resolve() / notebook_name)
output_path = output_dir / pdf_name

print(f' Generando PDF con Quarto...')
print(f' Archivo de salida: {output_path}')

result = subprocess.run(
    f'quarto render {notebook_abs} --to pdf --output-dir "{output_dir}" --output {pdf_name}',
    shell=True,
    capture_output=True,
    text=True
)

if result.returncode == 0:
    print(f' PDF generado exitosamente: {output_path}')
else:
    print(f' Error al generar PDF:')
    print(result.stderr)

```

```

Generando PDF con Quarto...
Archivo de salida: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standard\sae_p-
annealing_standard\5000games\ex2\metrics\sae_standard_p-annealing_16_5000g_ex2_reconstruction_absolute.pdf
PDF generado exitosamente: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standard\sae_p-
annealing_standard\5000games\ex2\metrics\sae_standard_p-annealing_16_5000g_ex2_reconstruction_absolute.pdf

```